

# A Novel Instantaneous Frequency Algorithm and Its Application in Stock Index Movement Prediction

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**Abstract**—There are different approaches in defining instantaneous frequency in the literature; however, with a great variety of practical problems a specific approach is not always applicable. This paper proposes a new type instantaneous frequency suitable for a large class of signals called simple wave. The new instantaneous frequency is called  $\alpha$ -counting instantaneous frequency (IF) that should have independent interest. A stock index movement prediction approach is proposed in this paper based on the newly defined instantaneous frequency and signal decomposition. An experiment is provided based on the stock market data of Hong Kong Hang Seng Index. Effectiveness of the proposed approach is assessed through the comparison with BP neural network prediction method, and is proved to be promising.

**Index Terms**— $\alpha$ -counting IF, backpropagation (BP) neural network, empirical mode decomposition (EMD), intrinsic mode function (IMF), instantaneous frequency (IF), radial basis function (RBF) neural network, stock price movement prediction.

## I. INTRODUCTION

IN many signal processing applications, it is often of significant importance to use the concept of instantaneous frequency (IF) to describe the frequency at a particular time, which represents the detailed mechanism and physical phenomenon of signals. IF stems from the fact that in practice frequencies of a signal vary with time. Due to theoretical limitation, some existing IF concepts cannot meet various practical demands [1]. Researchers have been making efforts in finding more practical, acceptable and workable ways to define and computer IF.

In the time-frequency literature, frequency is conditionally defined as the derivative of the phase of the signal. The frequency of a sinusoidal signal is a well-defined concept. In practice, however, signals are not truly sinusoidal but aggregates of sinusoidal components. For such signals, the notion of frequency loses its practical sense, and one needs to use a parameter which accounts for the time-varying nature of the process, and, accordingly, time-varying instantaneous frequencies may be defined [1], [2].

The physical meaning of frequency is defined as the number of vibrations per unit time. It should be a nonnegative quantity varying along with time. A signal which has no negative Fourier frequency components is called an *analytic signal*. There are various methods to define frequency of analytic signals. In the case, IF can be defined as the average Fourier frequency at each time. In general one cannot expect every function has well defined IF by any definition. With the traditional frequency analysis method—Fourier series and Fourier integral representations, one first defines it in the harmonic wave case and then expands a series of harmonic waves. To make the representation more local, one started with window Fourier transforms. Lately, wavelet transforms were studied, where frequencies correspond with the dilation parameters. However, these mentioned types of frequencies cannot give the effective time-varying frequency representations or instantaneous characteristics of signals. Traditionally, IF was computed from analytic signal. Constructing an analytic signal via the Hilbert transformation (HT) is equivalent to eliminating the negative Fourier spectrum [1]. There has been an ample amount of studies on IF based on analytic signal [1], [3].

In 1998 the so called empirical mode decomposition (EMD) algorithm was proposed by Huang *et al.* [4]. EMD provides a highly local and adaptive approximation via kind of basic functions called intrinsic mode functions (IMFs). An IMF has no mathematical definition and such approximation cannot be well estimated. There is only one method to verify whether a function is an IMF of some EMD that is to run the EMD algorithm to it: If the EMD cannot decompose the given function, then the given function is an IMF with respect to that particular EMD, and otherwise not. Within a tolerant threshold in computer one does obtain a sequence of IMFs for further treatment.

The EMD algorithm rests in the expectation that the analytic IFs obtained via the associated analytic signals of IMFs are non-negative [2], [5]. That is, if  $f$  is an IMF, then by expressing

$$f(t) + iHf(t) = \rho(t)e^{i\theta(t)}$$

One expects

$$\theta'(t) \geq 0. \quad (1)$$

Unfortunately, it is not the case. In [6], it is shown that there exist IMFs, obtained through EMD, that do not enjoy the property (1). Specifically, it means that IMFs obtained through EMD do not behave well with Hilbert transformation, and, in fact, it has no relation with Hilbert transform [6], [7]. A comprehensive theory with applications of analytic signals with nonnegative phase derivative, i.e., mono-components, is studied in [8], [9], and [10].

Financial data with different frequencies are important in studying a variety of issues related to the trading process

Manuscript received June 30, 2011; revised January 12, 2012; accepted May 02, 2012. Date of publication May 11, 2012; date of current version July 13, 2012. This work was supported by the University of Macau research grant MYRG144(Y1-L2)-FST11-ZLM. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Sanjeev Kulkarni.

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Digital Object Identifier 10.1109/JSTSP.2012.2199079

and market microstructure. They can be used to compare the efficiency of different trading systems in price discovery. They can also be used to study the dynamics of bid and ask quotes of a selected stock [11]. In particular, high-frequency data have some unique characteristics that do not appear in lower frequencies. The EMD algorithm is highly local and adaptive. In this situation, EMD algorithm can stand as one of the approaches for analysis of the financial data, but the produced IFMs suffer from having negative analytic IFs.

To solve the above problem, in this paper we propose a new concept of IF. It focuses on a class of signals called simple wave (SW) that contains IMFs as its proper subset. The proposed new IF concept uses a straightforward method with simple formula and algorithm to compute its value. It can become an important supplement to EMD algorithm and is suitable for engineering applications in high-frequency signal analysis.

Stock index movement prediction is one of the hot areas in financial data analysis. There have been a number of methods applied to this area, including the traditional statistical economics and artificial neural network. Limitation of traditional statistical economics consists of inflexibility in dynamic situation and complexity of modeling owing to the presence of noise and nonlinearity in the time series [12]. Backpropagation (BP) neural network based stock movement prediction approach is considered as a powerful tool for the nonlinear dynamic system prediction, which can better predict the short-term trend of the stock market index [12], [13]. In this paper, a new stock index movement prediction approach—SW signal decomposition based prediction is proposed. The  $\alpha$ -counting IF of SW is introduced and used to find out the primary, secondary and average cycles of the stock volatility in each of the composing IMFs. Besides the proposed application of  $\alpha$ -counting IF, because of its general applicability, the newly defined IF should have considerable and independent significance. Then radial basis function (RBF) neural network is combined into our method for the stock index movement prediction [14]. The performance between the proposed prediction approach and BP neural network prediction approach is compared. The result of our proposed method is found better.

The paper is organized as follows. The new mathematical definition of IF for SW and its computation algorithm are introduced in Section II. The principle of the proposed SW based stock index movement prediction is described in Section III. The experiment result is shown in Section IV. The conclusions are drawn in Section V.

## II. NEW MATHEMATICAL DEFINITION OF IF FOR SW

Based on the practical applications, an expected and qualified IF concept should satisfy the following three requirements.

- *Time varying property.*
- *Non-negativity.*
- *Reflecting the local frequency properties of the signal.*

In particular, the last condition would include the requirement that the IF of  $\cos(nt)$  should be  $n$ . The analytic signal method does not satisfy the second requirement [2], [5], [8], but satisfies the first and the third requirements.

Below we define the so-called  $\alpha$ -counting IF for a large class of basic signals SWs. The definition is straightforward and easy to apply. The  $\alpha$ -counting IF satisfies all the three mentioned requirements.

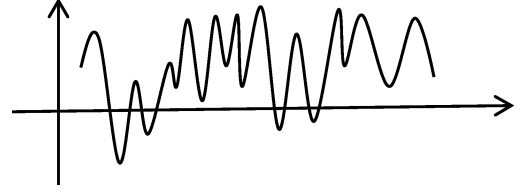


Fig. 1. Example of SW.

### A. Definitions of SWs and $\alpha$ -counting IF

**Definition 2.1:** We call a function defined on a finite or an infinite interval a simple wave (SW), if:

- 1) it is continuous with only strict local extrema;
- 2) between every adjacent pair of strict local maximum and minimum the function is monotone;
- 3) in any finite interval the function has only finitely many strict extrema and finitely many subintervals on which the function is constant.

An example of SW is illustrated in Fig. 1. By saying that  $f$  has a *strict local maximum* at  $t_0$  if

$$f(t_0 + \Delta t) < f(t_0)$$

for all sufficiently small  $\Delta t > 0$ . *Strict local minimum* is defined similarly.

By saying that  $f$  is *monotone in an interval* we mean

$$f(t + \Delta t) \geq f(t) \quad \text{for all } t \quad \text{and} \quad \Delta t > 0$$

or

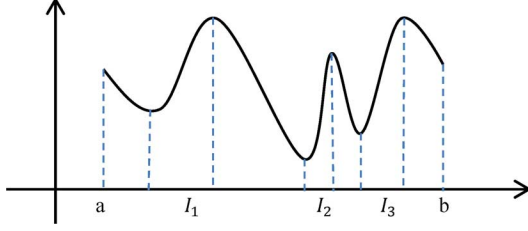
$$f(t + \Delta t) \leq f(t) \quad \text{for all } t \quad \text{and} \quad \Delta t > 0$$

where both  $t$  and  $t + \Delta t$  are in the interval. The above two cases are said to be, respectively, *monotonously increasing* and *monotonously decreasing*, or, in brief, *increasing* or *decreasing*. If, in the above two inequalities, instead of “ $\geq$ ” and “ $\leq$ ” we have “ $>$ ” and “ $<$ ”, respectively, then we say that in the two cases  $f$  is strictly increasing or strictly decreasing.

By saying that  $f$  is *strictly increasing or strictly decreasing on an open interval*  $(a, b)$ , if  $f$  is, respectively, increasing or decreasing in  $(a, b)$ , and  $f(a) > f(b)$  or  $f(a) < f(b)$ .

The new IF concept is based on the physical meaning of frequency being approximately the number of vibrations per unit time. The proposed method is to find the minimum period  $2\alpha$  of the vibration, regardless the amplitudes, and thus to find the corresponding frequency at the point.

Denote by  $\Omega$  the domain of a SW function  $f$ . Let  $(a, b) \subset \Omega$ , and  $I_1, \dots, I_l$  be a set of  $l$  nonoverlap open intervals in  $(a, b)$  ordered from left to right. If  $f$  is strictly increasing **on** every  $I_k$ ,  $k = 1, \dots, l$ , and between each pair  $I_k$  and  $I_{k+1}$  there exists an open interval **on** which the function  $f$  is strictly decreasing, then we say that the set of the nonoverlap intervals  $I_1, \dots, I_l$  is *l-increasingly separable* (Here we allow  $l = 1$ ). Let  $I^+(a, b)$  denote the largest cardinal number  $l$  of all possible sets of increasingly separable open intervals contained in  $(a, b)$ . Similarly one defines  $I^-(a, b)$  to be the largest cardinal number  $l$  of all possible sets of decreasingly separable open intervals in  $(a, b)$ . The properties of SW guarantee that  $I^+(a, b)$  and  $I^-(a, b)$  differ at most by 1. Denote


 Fig. 2. Nonoverlap open intervals in  $(a, b)$ .

$I(a, b) = \max\{I^+(a, b), I^-(a, b)\}$ . The above processing is illustrated in Fig. 2.

Now let  $f$  be a SW and  $t$  be any but fixed real number in the domain of  $f$ , and  $\alpha > 0$ . Define an integer-valued function (depending on  $f$ )

$$A_\alpha(t) := \left\lfloor \frac{1}{2} I(t - \alpha, t + \alpha) \right\rfloor \quad (2)$$

where for any real number  $x$ , the notation  $[x]$  stands for the maximal integer that does not exceed  $x$ . We have

**Theorem 2.1:** There exists  $\delta > 0$  such that  $A_\alpha(t) = 0$ , whenever  $\alpha < \delta$ .

*Proof:* Since in any finite interval there exist only finitely many extrema, one can always choose a small enough  $\delta > 0$  such that for any  $\alpha < \delta$  the interval  $(t - \alpha, t + \alpha)$  contains at most one extremum point of  $f$ . For such  $\alpha$  we have  $A_\alpha(t) = 0$ . ■

Based on Theorem 2.1 we have

$$\delta(t) = \sup\{\alpha : A_\alpha(t) = 0\} > 0. \quad (3)$$

Define

$$\theta'(t) = \frac{\pi}{\delta(t)}. \quad (4)$$

Then  $\theta'$  is called  $\alpha$ -counting instantaneous frequency, abbreviated as  $\alpha$ -counting IF.

We note that the notation  $\theta'$  should not be understood as derivative as there is no definition for  $\theta$ . We provide the reasoning of  $\alpha$ -counting IF as follows. If  $\alpha$  is small enough, then  $A_\alpha(t) = 0$ . When  $\alpha$  grows larger and larger, it goes through certain values that make  $A_\alpha(t) = 1$ . The latter means that the larger one between  $I^+$  and  $I^-$  is 2 or 3. In the case the signal has 2 to 4 local extrema in the open interval  $(t - \alpha, t + \alpha)$ . In this situation we say that the signal has a vibration. By definition  $\delta(t)$  is the supremum of all the  $\alpha$  of which  $A_\alpha(t) = 0$ . One vibration takes  $2\delta(t)$  duration, in the  $2\pi$  duration it has altogether  $(2\pi)/(2\delta(t))$  vibrations, that is the value of the instantaneous IF.

The above defined is a type of instantaneous frequency. First, it is time varying. Secondly, it is nonnegative. Third, it represents the density degree of vibrations at the time moment  $t$ .

In principle, the above defined  $\alpha$ -counting IF can be used. From the above analysis, however, it is not so accurate. For  $f(t) = \cos nt$  the above defined  $\delta(t)$  does not give rise to the constant frequency  $n$ . Mathematically, in the case it would be desirable that  $\theta'(t) = n$ . By doing a little modification on the above defined  $A_\alpha$  we can achieve this. Although it increases the complication of the mathematics and the computation, it gives

certain consistency with the traditional concept. We will adopt the notation as follows:

$$\begin{cases} I_\alpha^+(t) = I^+(t - \alpha, t + \alpha), \\ I_\alpha^-(t) = I^-(t - \alpha, t + \alpha) \\ I_\alpha(t) = \max\{I_\alpha^+(t), I_\alpha^-(t)\} \end{cases}.$$

Define

$$\tilde{A}_\alpha(t) = \left\lfloor \frac{1}{3} (I_\alpha^+ + I_\alpha^- - 1 + \chi_-(f'(t - \alpha)(f(t - \alpha) - f(t + \alpha)))) \right\rfloor \quad (5)$$

where  $\chi_-$  is the characteristic function of the set  $(-\infty, 0]$ , i.e.,

$$\begin{aligned} \chi_-(t) &= 1, & t &\in (-\infty, 0] \\ \chi_-(t) &= 0, & t &\in (0, \infty). \end{aligned}$$

With  $\delta(t)$  defined as in (3) but with the modified  $\tilde{A}_\alpha$ , we have the following theorem.

**Theorem 2.2:** The above defined IF  $\theta'(t)$  is a continuous and positive function in  $t$ , and with  $f(t) = \cos nt$  there holds  $\theta'(t) = n$ .

*Proof:* We first show the continuity. The quantity  $I_\alpha^+ + I_\alpha^- - 1 + \chi_-(f'(t - \alpha)(f(t - \alpha) - f(t + \alpha)))$  in the definition of  $\tilde{A}_\alpha(t)$  is, in fact, a measurement of when there exist more than zero periods in  $(t - \alpha, t + \alpha)$ : it is when the above quantity have a value large or equal 3, meaning that we have totally 2 or more than 2 local extrema points. In the former case, from  $f(t - \alpha)$  to  $f(t + \alpha)$  the function undergoes, in a sense, a full cycle. In fact, we use this formula to define when we have a period. Let  $t$  be given and  $\delta(t)$  be defined through  $\tilde{A}_\alpha(t)$  in (3). Now consider a small movement of  $t$  to  $t + \Delta t$ ,  $\Delta t > 0$ . Then

$$\begin{aligned} (t + \Delta t - (\delta(t) - \Delta t), t + \Delta t + (\delta(t) - \Delta t)) \\ \subseteq (t - \delta(t), t + \delta(t)). \end{aligned}$$

This shows that  $\delta(t + \Delta t) \geq \delta(t) - \Delta t$ . On the other hand,

$$\begin{aligned} (t + \Delta t - (\delta(t) + \Delta t), t + \Delta t + (\delta(t) + \Delta t)) \\ \supseteq (t - \delta(t), t + \delta(t)). \end{aligned}$$

This shows that  $\delta(t + \Delta t) \leq \delta(t) + \Delta t$ . Altogether,

$$|\delta(t + \Delta t) - \delta(t)| \leq \Delta t.$$

Similarly, for  $\Delta t < 0$ , we can show

$$|\delta(t + \Delta t) - \delta(t)| \leq -\Delta t.$$

Therefore,

$$|\delta(t + \Delta t) - \delta(t)| \leq |\Delta t|.$$

This is the continuity.

Next we prove the cosine related property. If  $\alpha < \pi/n$  and  $t' \in (t - \alpha, t + \alpha)$ , then  $nt' \in (t - n\alpha, t + n\alpha)$ , where the length of the interval  $2n\alpha < 2\pi$ .

For all the possible cases,  $I_\alpha^+ + I_\alpha^- - 1$  is one or two; and when  $I_\alpha^+ + I_\alpha^- - 1$  is two,

$$\chi_-(f'(t - \alpha)(f(t - \alpha) - f(t + \alpha))) = 0.$$

So, in all the cases  $\tilde{A}_\alpha(t) = 0$ .

Next let  $\alpha > \pi/n$ , and thus  $nt' \in (t - n\alpha, t + n\alpha)$ , where the length of the interval is  $2n\alpha > 2\pi$ , larger than a period of cosine. We show that  $\tilde{A}_\alpha \geq 1$ . In the case  $I_\alpha^+ + I_\alpha^- \geq 3$ . If  $I_\alpha^+ + I_\alpha^- > 3$ , we are done. We found that for all the possibilities for  $I_\alpha^+ + I_\alpha^- = 3$ , there holds

$$\chi_-(f'(t - \alpha)(f(t - \alpha) - f(t + \alpha))) = 1.$$

Therefore,  $\tilde{A}_\alpha = 1$ . We thus conclude  $\delta(t) = \pi/n$ , and thus  $\theta'(t) = n$ . ■

The formula (3)(5) are used to compute the projected cycle days in applying the neural network program for stock index analysis.

### B. IF Computation Algorithm

Based on the above mathematical definition, the defined IF applies only to the SWs. One way to obtain SWs is through the EMD decomposition. Based on EMD algorithm, Each IMF satisfies the following two properties.

- 1) Between every adjacent pair of local extrema there is a zero crossing.
- 2) The average of upper envelop and lower envelop is zero.

An example of IMF is shown in Fig. 3. From the above IMF properties, one can see that IMFs are special case of SWs. The detailed  $\alpha$ -counting IF computation algorithm is illustrated in Algorithm 2.1.

#### Algorithm 2.1 Algorithm of $\alpha$ -counting IF Computation

**Input:**  $f(t)$  (a SW signal in  $[a, b]$ );

**Output:** IF;

Find all the extremas of  $f(t)$ ;

$t = a$ ;

**while**  $t \leq b$  **do**

$\alpha = 1$ ;

**while**  $\alpha \leq \min(ab)$  **do**

$I_\alpha^+$  = number of increasing intervals in  $(t - \alpha, t + \alpha)$ ;

$I_\alpha^-$  = number of decreasing intervals in  $(t - \alpha, t + \alpha)$ ;

**if**  $f'(t - \alpha)(f(t - \alpha) - f(t + \alpha)) \leq 0$  **then**

$$\tilde{A}_\alpha(t) = \text{fix}(1/3(I_\alpha^+ + I_\alpha^-));$$

**else**

$$\tilde{A}_\alpha(t) = \text{fix}(1/3(I_\alpha^+ + I_\alpha^- - 1));$$

**end if**

**if**  $\tilde{A}_\alpha(t) > 0$  **then**

$$\delta(t) = \alpha;$$

$$IF(t) = \frac{\pi}{\delta(t)};$$

$$t = t + 1;$$

**break**;

**else**

$$\alpha = \alpha + 1;$$

**end if**

**end while**

**end while**

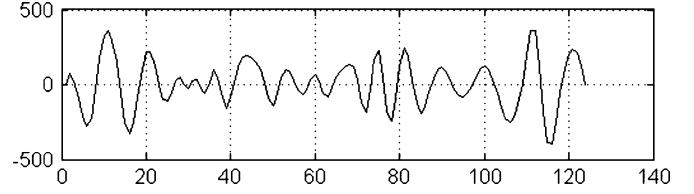


Fig. 3. Example of IMF.

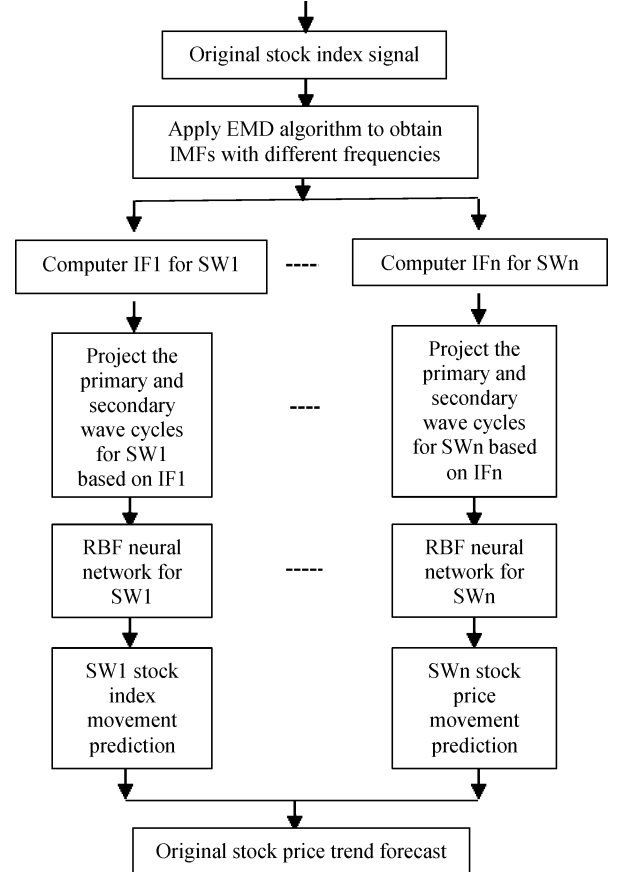


Fig. 4. Flowchart of SW-based stock price movement prediction.

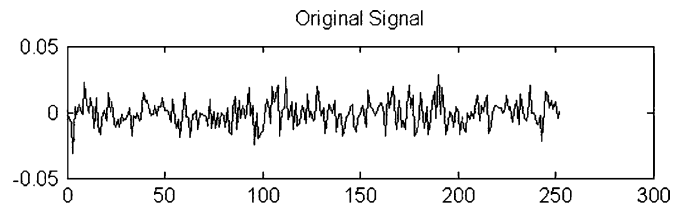


Fig. 5. Daily returns of Hong Kong's Hang Seng Index (16/06/2010–14/06/2011).

### III. PRINCIPLE OF THE SW-BASED STOCK INDEX MOVEMENT PREDICTION APPROACH

The stock market is a complex system. Over the years, people have made unremitting studies in predicting the stock index movements. There are a number of prediction methods. BP neural network algorithm is one of the effective ones. However, the convergence of BP algorithm is slow and the initial parameter data selection remains as an art rather than a science [12]. In this paper, a new prediction approach is proposed, which is

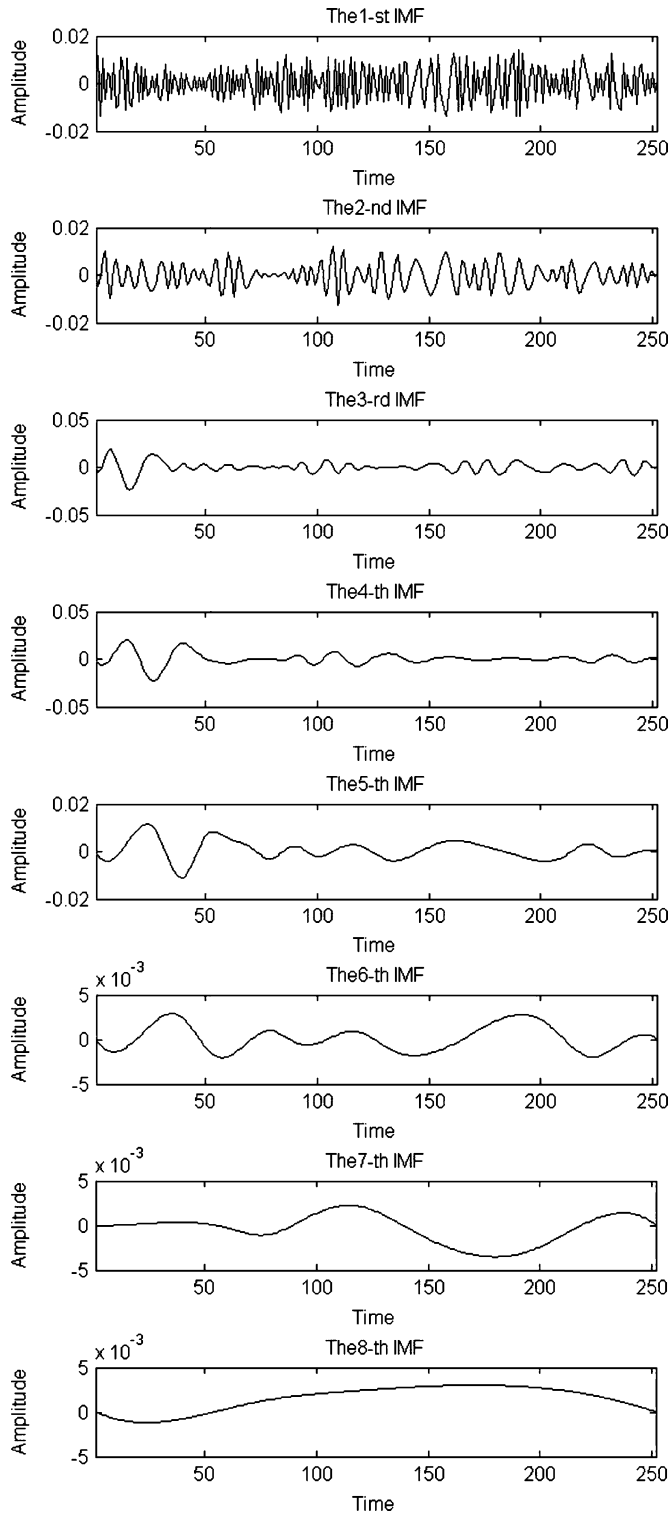


Fig. 6. IMFs decomposed by EMD.

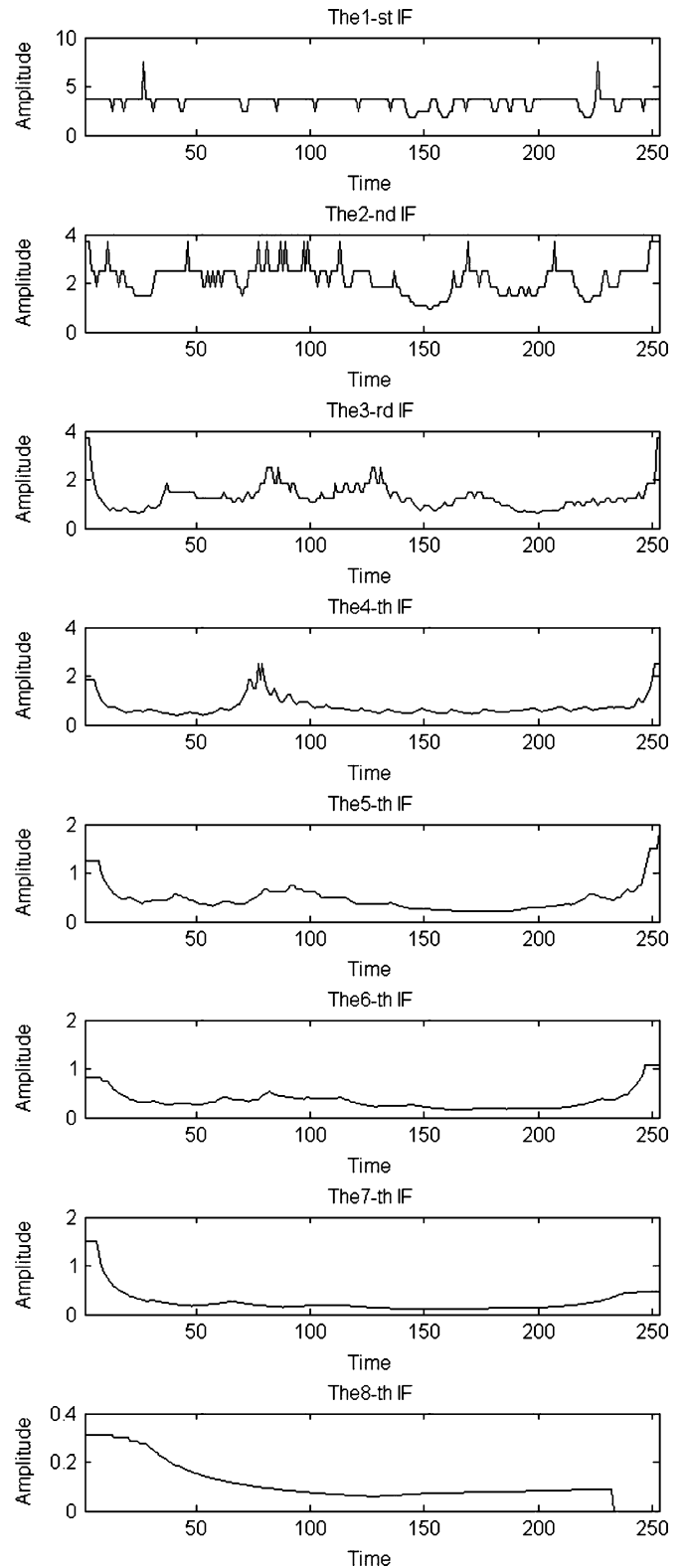


Fig. 7. IFs of the SWs.

based on the idea of the signal decomposition and  $\alpha$ -counting IF defined for each composing SW.

The principle of the proposed SW based stock index movement prediction approach is as follow. The complex stock index signal is first decomposed into IMFs through EDM algorithm. The IMFs are listed based on their frequencies, from higher frequencies to lower frequencies. The IF is computed and used

in wave cycle projection for each SW. Then the RBF neural network is used to predict stock index movement for each SW taking the projected wave cycles as the input values. At last, all prediction results are combined to form the original stock index trend forecast.

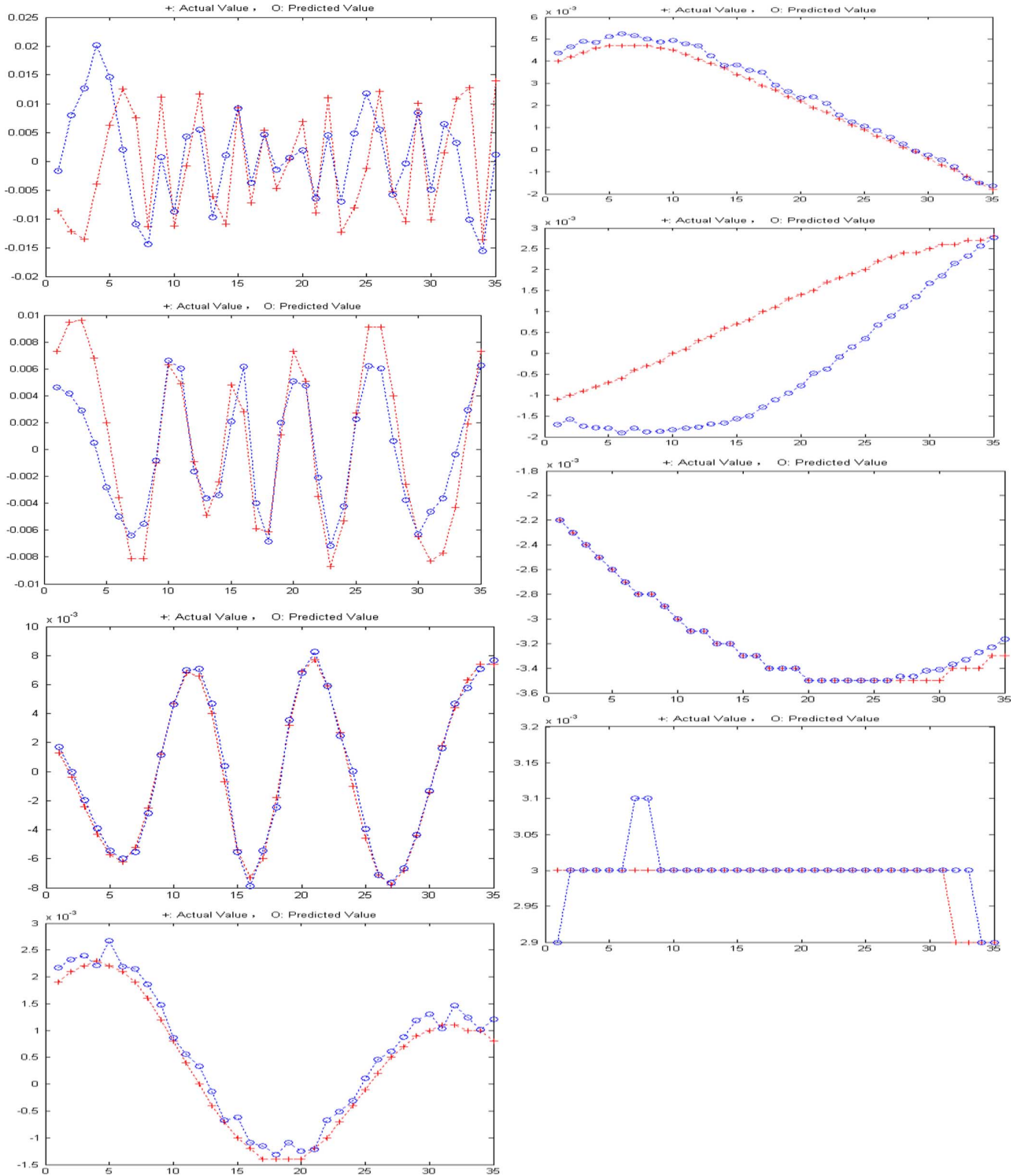


Fig. 8. RBF prediction results for each component IMFs.

#### IV. EXPERIMENT RESULTS

##### A. Experiment Design

The experiment data is selected from one year closing price of Hong Kong's Hang Seng Index for the period from 15th June, 2010 to 14th June, 2011. There were total 253 stock market trading days during the selected one year. The daily returns of the period 16th June, 2010 to 14th June, 2011 (total 252 trading

days) form the original signal. The daily returns are calculated by the formula as follows:

$$\text{Day 2 daily return} = \frac{(\text{Day 2 closing price} - \text{Day 1 closing price})}{\text{Day 1 closing price}}.$$

The original signal is shown in Fig. 5. The flowchart of the proposed approach is illustrated in Fig. 4.

The EMD algorithm is applied first to decompose the original signal into its composing IMFs. The decomposition results are shown in Fig. 6. There are eight IMFs decomposed by EMD algorithm corresponding to the original data. Then eight IFs' functions are computed based on the Algorithm 2.1. The corresponding IFs are illustrated in Fig. 7.

The primary and secondary cycles of each SW are projected based on the value of IFs. The primary cycle is classified as the duration period of IF that comes most often, and the secondary cycle is the duration period of IF that comes least. The duration period is corresponding to the value of  $2\alpha$  in the algorithm. Based on the data, the primary cycle of the component SW1 is quasi-2, and quasi-4 days, and the secondary cycle is quasi-5, quasi-9, and quasi-12, etc. For those longer cycles, however, the number of occurrences is only once. Similarly, the primary cycles of SW2-SW7 components are 4 and 5 days, 14 and 15 days, 25 and 26 days, 45, 46 and 48 days, 75 days, and 150 days, respectively.

SW8 component reflects the overall trend of the daily returns in the selected period.

Three-layer RBF neural network structure is selected as the prediction model in this study, including an input layer, a hidden layer and an output layer. In the input layer of SW1-SW8, the input nodes are determined based on the projected cycles in last step. They are 4, 5, 15, 25, 45, 75, 150, and 10, respectively.

The hidden layer nodes can select automatically by the software. The node of the output layer is 1. The Matlab RBF neural networking function *newrb()* is used in the experiment, where spread = 100, and goal =  $1 \times 10^{-12}$ . Other default parameters are used directly in the experiment.

The experiment procedure is a training and test process. The daily return data from 18th June, 2010 to 17th January, 2011 (total 150 trading days) are used as the training set. The daily return data from 20th January, 2011 to 11th March, 2011 (total 35 trading days) are used for testing the trend prediction.

### B. Experiment Results Analysis

The daily return time series can be decomposed into seven components with different wave cycles and a trend component, which can reflect the complex nature of the multilevel and multi-time scale in the closing price changes of the Hong Kong Hang Seng Index. The trend of the daily returns reflects a rise trend of Hong Kong's Hang Seng index from mid-June to the end of December 2010, and the gentle downward trend from early January to mid-June 2011.

In the decomposed daily return time series, the first IMF provides the maximum amplitude, the highest frequency, and the shortest wavelength. The amplitude and frequency decrease, the wavelength increases through the components IMF2-IMF8.

The fluctuation changes in components IMF1 to IMF3 present a significant long-term memory, i.e., that a more substantial fluctuation existed in the past means that there would be a larger fluctuation trend in the future. Meanwhile, that a smaller fluctuation existed in the past means that there would be a small fluctuation trend in the future, unless some key events might happen that result in changing the trend.

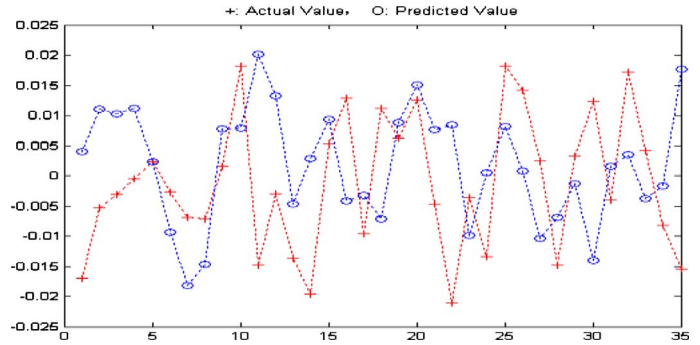


Fig. 9. Final RBF prediction results for the original daily return signal.

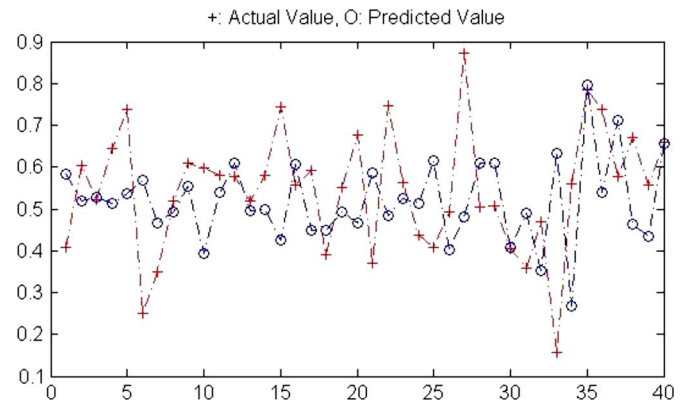


Fig. 10. Final BP prediction results for the original daily return signal.

The prediction results for each component IMFs and the original signal are illustrated in Figs. 8 and 9, respectively.

The error sums of the squares for each IMFs are 0.0034, 2.7185e-004, 5.9205e-006, 2.6020e-007, 2.6156e-007, 8.9921e-005, 9.4024e-007, and 3.8225e-008, respectively. The error sum of squares of the final prediction results is 0.0087.

In Fig. 8, we can see that besides IMF6, the predicted values can represent the actual movements' trends in every component. Furthermore, in IMF3-IMF5, and IMF7, the predicted values are very close to the actual ones. Those components can be used for further prediction studies, such as the different movements of the different types of traders. In Fig. 9, the synthesized predicted values can basically forecast the stock index movements of the original signal.

### C. Comparison With BP Neural Network Approach

To test the effectiveness of the proposed stock price prediction approach, BP neural network based prediction method is also applied to the same training and test data. The original signal is used directly for the forecast. The prediction result is shown in Fig. 10. The Matlab BP neural network function *newff()* is used in the experiment. The selected parameters are as follows. The number of the hidden nodes is 7. The time interval is 200. The training times are 300. Other default parameters are used directly in the experiment. As the results, the error sum of squares of this approach is 1.2560. Thus, our proposed approach significantly improves the prediction accuracy.

In our proposed approach, the stock price data are decomposed into different components with different frequencies.

Usually the high-frequency components are the system disturbance caused by noise, such as disordered movements of the retail capitals. The low-frequency components are often the stable output of the system, which represents the mainstream of the stock market capital movements and is more suitable in the market trends prediction. The disciplines of the above two kinds of components are different. In our proposed approach, the different values of the primary cycles were selected as the input nodes towards the decomposed different signal components in the prediction model. The forecast results were then synthesized on top of the different components. Therefore, the effect is better than that of the direct prediction approach.

## V. CONCLUSION

A novel instantaneous frequency called  $\alpha$ -counting IF is proposed. The newly defined IF satisfies the three requirements for instantaneous frequency. The principle is simple and straight forward. It can be used to simple wave, including IMFs obtained from an EMD algorithm. Its conceptual primitiveness and wide applicability make it to be of independent interest.

As an important application the proposed  $\alpha$ -counting IF is them used to analyze stock index following an application of EMD decomposition. It is predicted that the  $\alpha$ -counting IF method may have significant applications in financial data analysis.

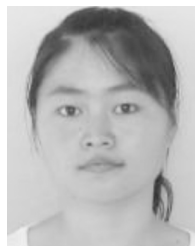
## REFERENCES

- [1] B. Boashash, "Estimating and interpreting the instantaneous frequency of a signal—Part 1: Fundamentals," *Proc. IEEE*, vol. 80, no. 4, pp. 520–538, Apr. 1992.
- [2] L. Cohen, *Time-Frequency Analysis: Theory and Application*. Englewood Cliffs, NJ: Prentice-Hall, 1995.
- [3] B. Boashash, "Estimating and interpreting the instantaneous frequency of a signal—Part 2: Algorithms and applications," *Proc. IEEE*, vol. 80, no. 4, pp. 540–568, Apr. 1992.
- [4] N. Huang, Z. Shen, S. R. Long, M. C. Wu, H. H. Shih, Q. Zheng, N. Yen, C. C. Tung, and H. H. Liu, "The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis," in *Proc. R. Soc. London, A*, 1998, vol. 454, pp. 903–995.
- [5] B. Picinbono, "On instantaneous amplitude and phase of signals," *IEEE Trans. Signal Process.*, vol. 45, no. 3, pp. 552–560, Mar. 1997.
- [6] R. C. Sharple and V. Vatchev, "Analysis of intrinsic mode functions," *Constr. Approx.*, vol. 24, pp. 17–47, 2006.
- [7] T. Qian, L. Zhang, and H. Li, "Mono-components in signal decomposition," *Int. J. Wavelets. Multires. and Inf. Process.*, vol. 6, no. 3, pp. 353–374, 2008.
- [8] T. Qian, "Mono-components for decomposition of signals," *Math. Meth. Appl. Sci.*, vol. 29, pp. 1187–1198, 2006.
- [9] T. Qian and Y. B. Wang, "Adaptive decomposition into basic signals of non-negative instantaneous frequencies—A variation and realization of greedy algorithm," *Adv. Comput. Math. (AiCM)*, vol. 34, no. 3, pp. 279–293, 2011.
- [10] W. Mi and T. Qian, "Frequency domain identification with adaptive rational orthogonal system," in *Proc. IEEE Int. Conf. Syst. Sci. Eng. (IC SSE)*, 2010, pp. 41–45.
- [11] R. S. Tsay, *Analysis of Financial Time Series*, ser. Probability and Statistics, 2nd ed. New York: Wiley, 2005.
- [12] M. Thenmozhi, "Forecasting stock index returns using neural networks," *Delhi Bus. Rev.*, vol. 7, no. 2, pp. 59–69, 2006.
- [13] M. Lam, "Neural network techniques for financial performance prediction: Integrating fundamental and technical analysis," *Decision Support Syst.—Spec. Iss.: Data Mining for Financial Decision Making*, vol. 37, no. 4, pp. 567–581, 2004.
- [14] Y. Sun, Y. Liang, W. Zhang, H. Lee, W. Lin, and L. Cao, "Optimal partition algorithm of the RBF neural network and its application to financial time series forecasting," *Neural Comput. Applicat.*, vol. 14, pp. 36–44, 2005.



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